



Network Science Institute
at Northeastern University

N Northeastern University
London



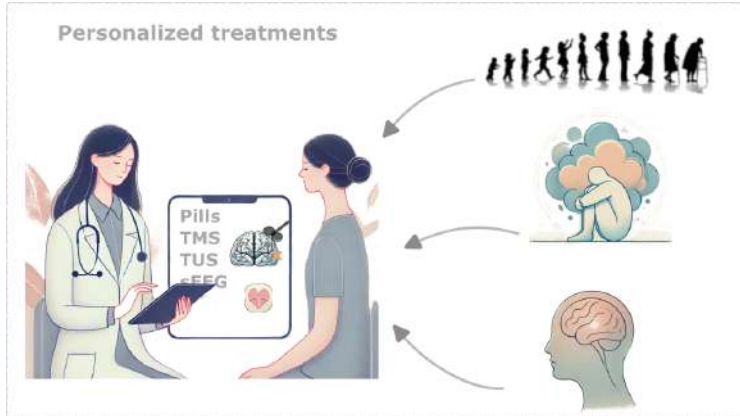
Higher-order analysis in neuroscience

Marilyn Gatica

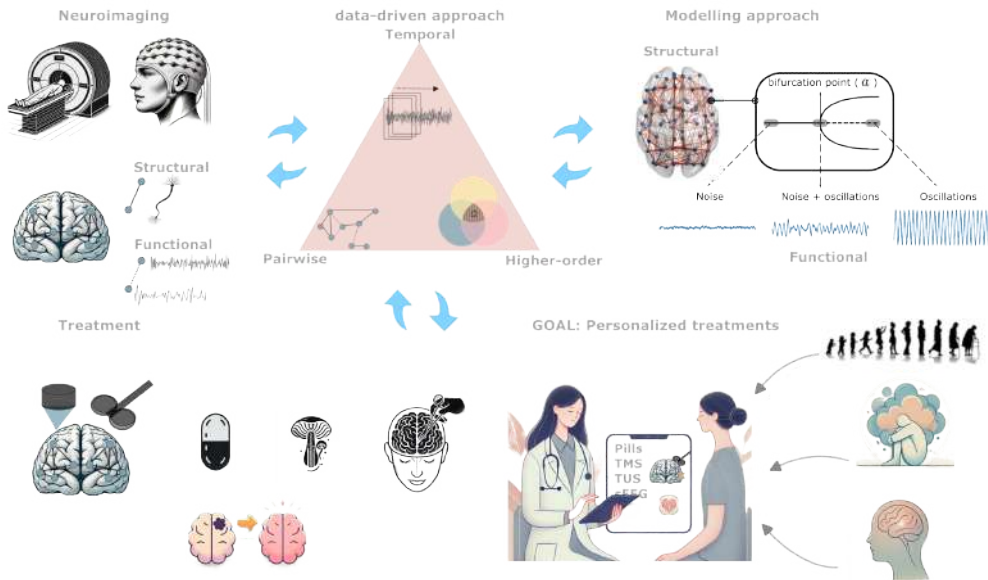
Postdoctoral Research Assistant in the Network Science Institute, Northeastern University London



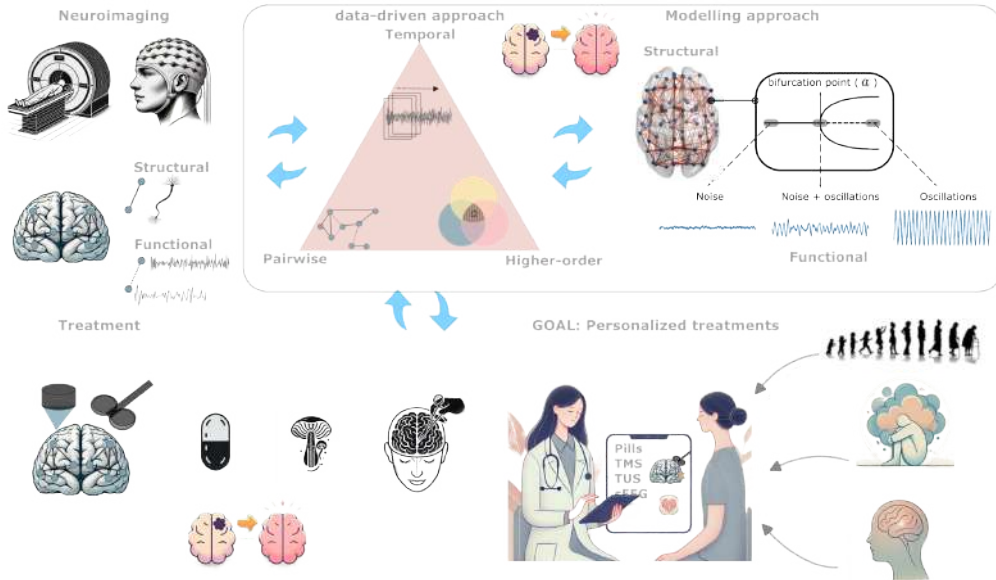
- Motivation
- Higher-Order data-driven analysis in computational neuroscience
- Charting Higher-Order models of brain function
- Transcranial ultrasound stimulation



Motivation

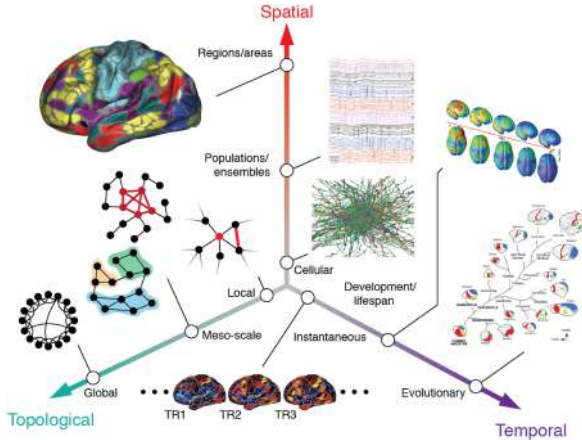


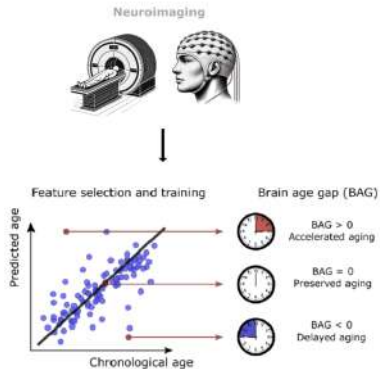
Motivation



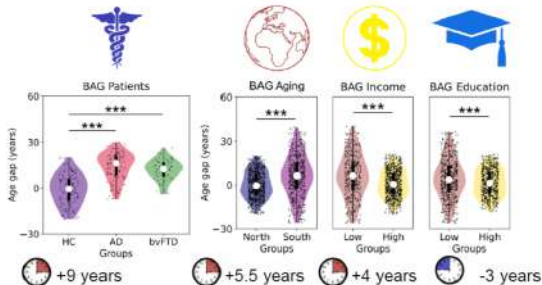
Challenges

How do we move forward toward our goal, given the complexity across multiple scales?



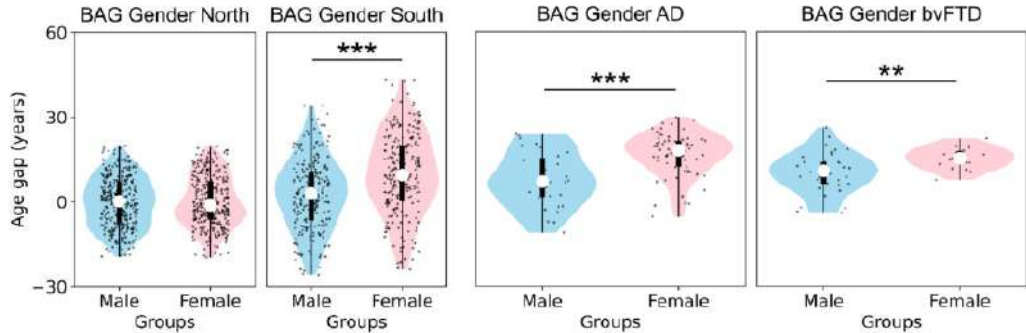


BAGs in healthy aging and dementia: accelerated aging in South and in dementia, and protective effects of education



Coronel-Oliveros et al., 2025. Accepted in Nature Mental Health.

Country-level gender inequality



Country-level gender inequality



AUTHOREA

An integrated computational framework for diversity-sensitive personalized medicine

BRAIN HEALTH DEMENTIA DIVERSITY GLOBAL SOUTH HIGH-ORDER INTERACTIONS
PERSONALIZED MEDICINE WHOLE-BRAIN MODELING

 +1 Carlos Coronel-Oliveros , Marilyn Gatica, Rubén Herzog, Matteo Neri 

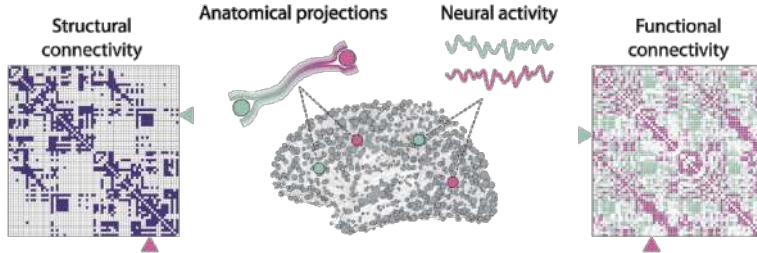


A B S T R A C T

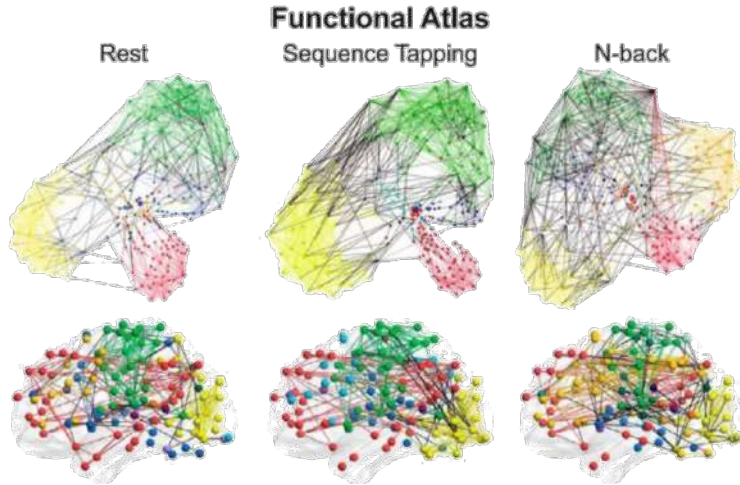
Diversity in biological, social, and environmental factors plays a central role in shaping brain health and disease. Distinct brain disorders frequently exhibit overlapping clinical phenotypes, despite arising from heterogeneous biological and contextual mechanisms. This convergence challenges conventional, population-averaged approaches, which often fail to capture interindividual variability and lead to limited reproducibility, weak translational potential, and inadequate tools for individual-level characterization. To address these gaps, we propose an integrative computational approach that unites normative models of brain aging (“brain clocks”), high-order interactions, and whole-brain modeling. Brain clocks estimate individualized brain health scores by comparing observed brain features to normative age-based trajectories. Brain high-order interactions capture functional dependencies beyond pairwise connectivity, offering sensitive biomarkers that reflect system-level diversity in aging and neurodegeneration. Whole-brain modeling uses theory-based simulations of individual brain dynamics, supporting the inference of latent mechanisms and the evaluation of targeted perturbations *in silico*. Together, they form a synergistic approach: normative models provide personalized baselines, high-order interactions enhance sensitivity to complex alterations, and whole-brain simulations enable causal insight and guide potential interventions. By embedding inter-individual variability and contextual diversity into each computational layer, this framework moves the field toward precision neuroscience, where assessment, understanding, and treatment are tailored to the individuals’ unique biological and social profiles.

- Our main goal is to advance into **personalized treatments**.
- **New methodologies** and biomarkers are needed.
- **Mechanistic models** are essential.

Functional and structural connectivity

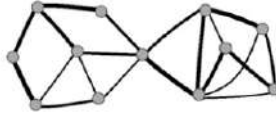
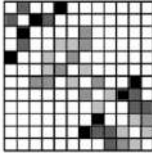


Suarez, et al 2020, Trends in Cognitive sciences



Brain Connectivity Toolbox (BCT)

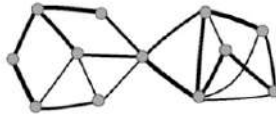
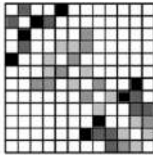
weighted undirected networks



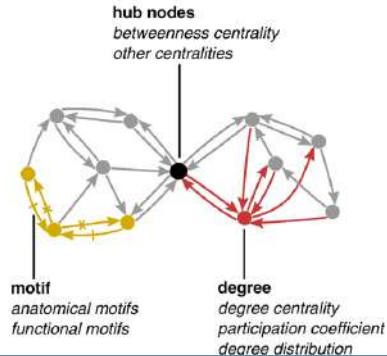
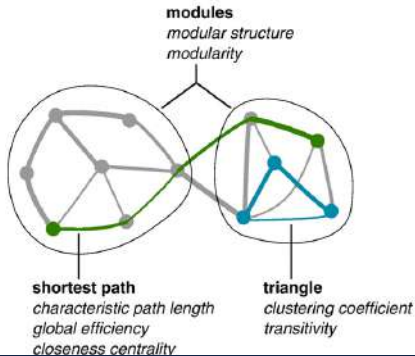
Rubinov M, Sporns O (2010) NeuroImage

Brain Connectivity Toolbox (BCT)

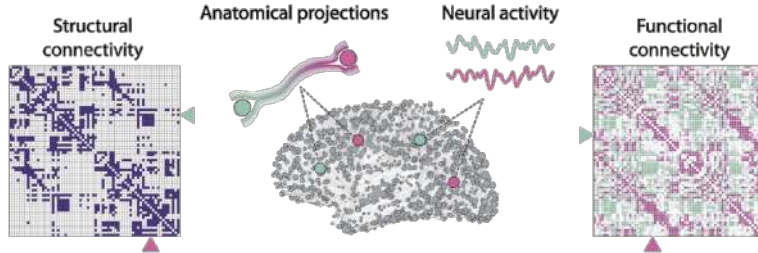
weighted undirected networks



Rubinov M, Sporns O (2010) NeuroImage

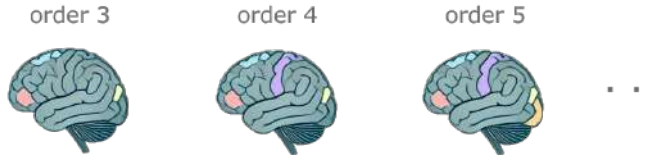


Beyond pairwise connectivity

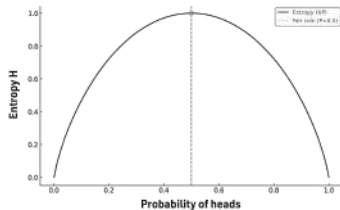


Suarez, et al 2020, Trends in Cognitive sciences

Higher-order interactions



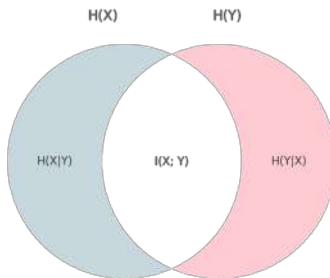
Entropy (n = 1)



$$H(X) = -\sum_{x \in \mathcal{X}} P(x) \log_2 P(x)$$

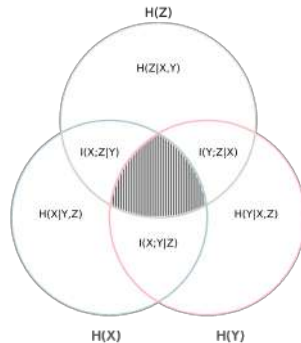
$$h(X) = -\int_{\mathcal{X}} f(x) \log_2 f(x) dx$$

Mutual Information (n = 2)



$$\begin{aligned} I(X; Y) &\equiv H(X) - H(X | Y) \\ &\equiv H(Y) - H(Y | X) \\ &\equiv H(X) + H(Y) - H(X, Y) \\ &\equiv H(X, Y) - H(X | Y) - H(Y | X) \end{aligned}$$

Interaction Information (n=3)



$$I(X; Y; Z) = I(X; Y) - I(X; Y | Z)$$

$$I(X; Y; Z) > 0 \text{ Redundancy}$$

$$I(X; Y; Z) < 0 \text{ Synergy}$$

High-order interactions ($n \geq 3$)



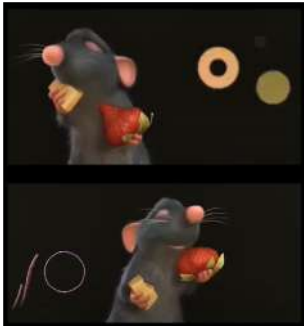
0-Information ($n \geq 3$)

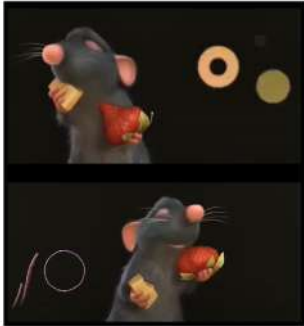
$$\Omega(X^n) := (n-2)H(X^n) + \sum_{j=1}^n [H(X_j) - H(X_{-j}^n)]$$

$\Omega(X^n) > 0$ redundancy-dominated

$\Omega(X^n) < 0$ synergy-dominated

Rosas, Mediano, et. al, 2019, Phys Rev X





Redundancy



Synergy



Redundancy



Redundancy

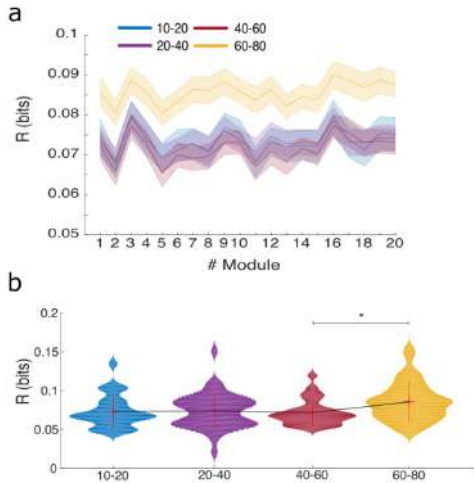


Synergy



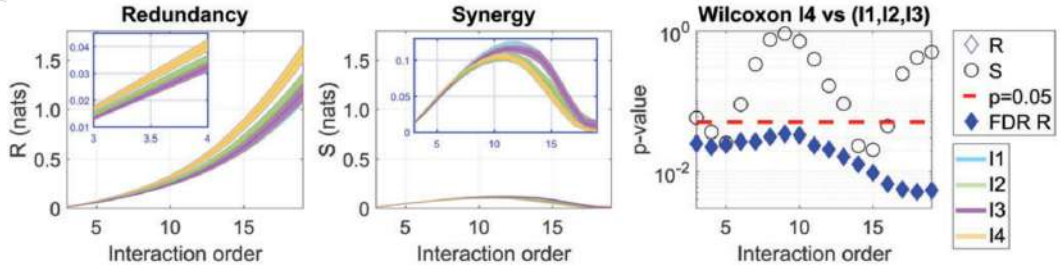
The extent of Higher-Order Interactions in Neuroscience





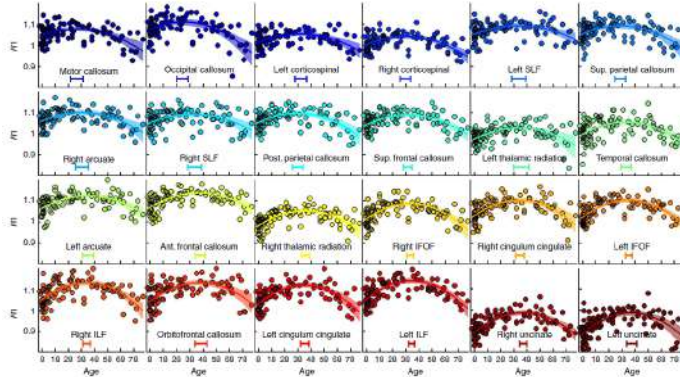
Camino-Pontes 2018, Entropy

Healthy aging



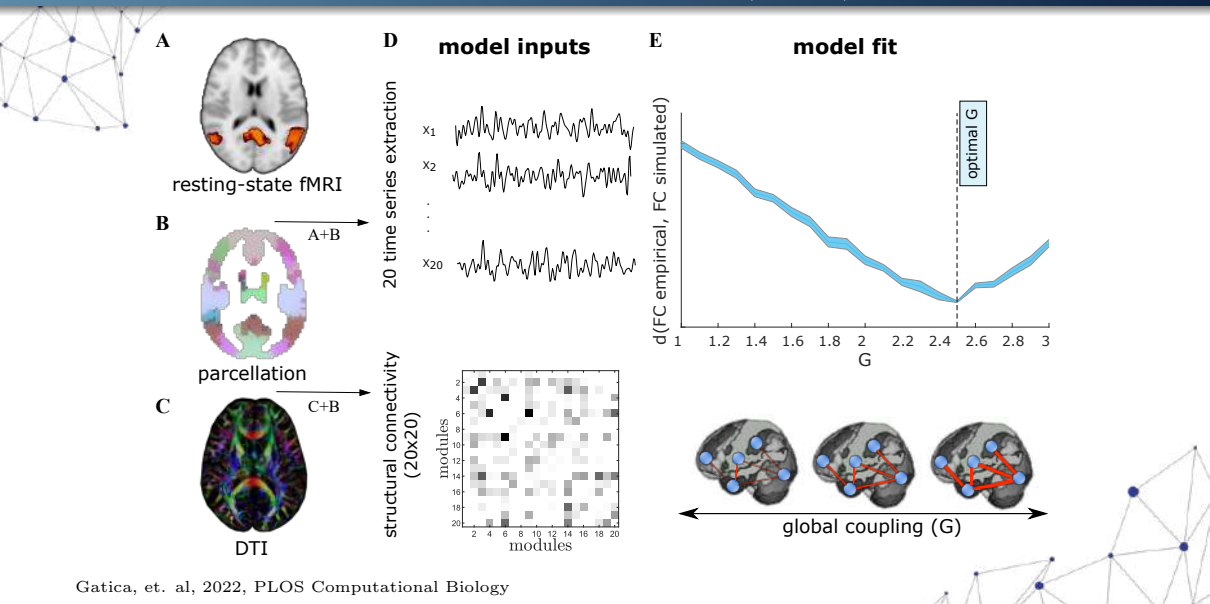
Gatica 2021, Brain Connectivity

Healthy aging: Inverted U-shape



Yeatman 2014, Nature Communications

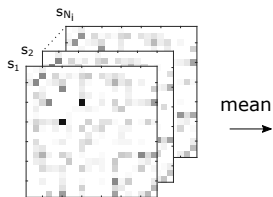
Dynamic mean field whole-brain modeling (DMF)





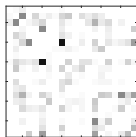
A

Connectomes group Ii

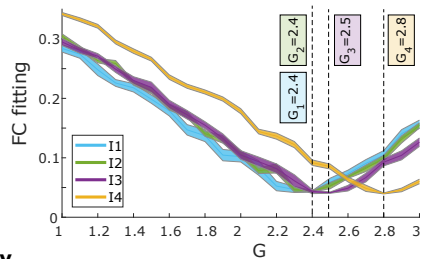


mean
→

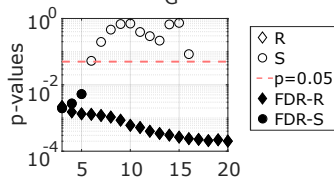
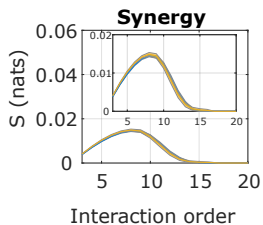
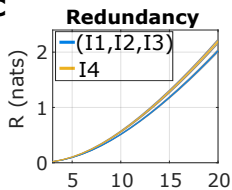
SC_i



B



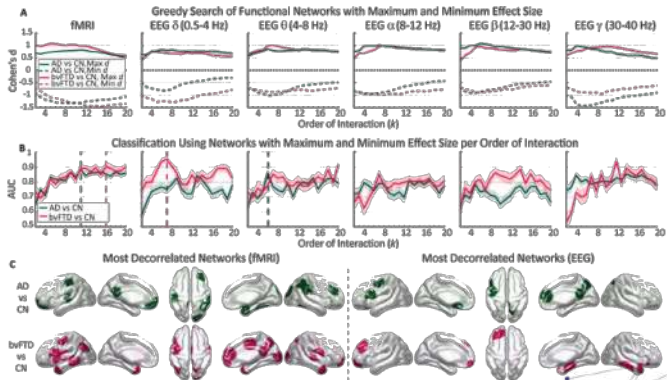
C





Genuine high-order interactions in brain networks and neurodegeneration

Rubén Herzog^{a,b}, Fernando E. Rosas^{b,c,d,e,f}, Robert Whelan^g, Sol Fittipaldi^{a,g,h},
Hernando Santamaria-García^a, Josephine Cruzat^{a,b}, Agustina Birbaⁱ,
Sebastian Moquilner^a, Enzo Tagliazucchi^{a,h}, Pavel Prado^a ,
Agustin Ibanez^{a,g,i,j}

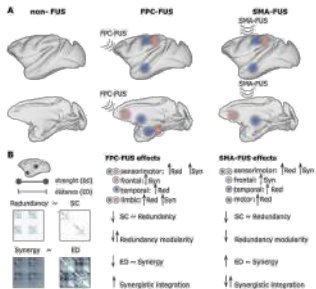


December 10 2024

Transcranial ultrasound stimulation effect in the redundant and synergistic networks consistent across macaques

In Special Collection: CogNet

Marilyn Gatica, Cyril Atkinson-Clement, Pedro A. M. Mediano, Mohammad Alkhawashki, James Ross, Jérôme Sallet, Marcus Kaiser

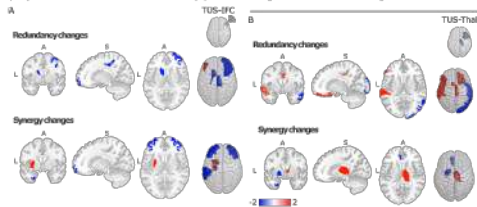


Understanding the high-order network plasticity mechanisms of ultrasound neuromodulation

Marilyn Gatica, Cyril Atkinson-Clement, Carlos Coronel-Oliveros, Mohammad Alkhawashki, Pedro A. M. Mediano, Enzo Tagliazucchi, Fernando E. Rosas, Marcus Kaiser, Giovanni Petri

doi: <https://doi.org/10.1101/2025.01.11.632528>

This article is a preprint and has not been certified by peer review [what does this mean?].

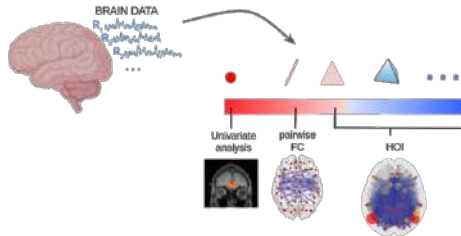


Review

> Eur J Neurosci. 2025 Feb;51(3):e16676. doi: 10.1111/ejn.16676.

A Taxonomy of Neuroscientific Strategies Based on Interaction Orders

Matteo Neri ¹, Andrea Brovelli ¹, Sammy Castro ^{2 3}, Fausto Fraioli ⁴, Marilyn Gatica ⁵,
Ruben Herzog ⁶, Pedro A M Mediano ^{7 8}, Ivan Mindlin ^{6 9}, Giovanni Petri ^{5 10 11},
Daniel Bor ^{12 13}, Fernando E Rosas ^{14 15 16 17}, Antonella Tramacere ¹⁸, Mar Estarellas ^{12 13}



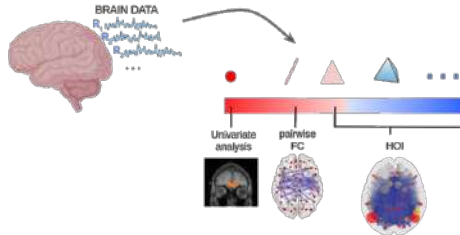
Review

> Eur J Neurosci. 2025 Feb;61(3):e16676. doi: 10.1111/ejn.16676.

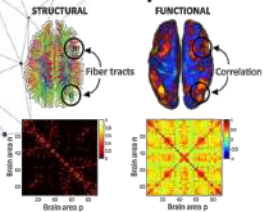


A Taxonomy of Neuroscientific Strategies Based on Interaction Orders

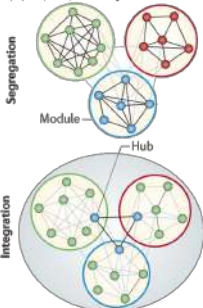
Matteo Neri ¹, Andrea Brovelli ¹, Sammy Castro ^{2 3}, Fausto Fraioli ⁴, Marilyn Gatica ⁵,
Ruben Herzog ⁶, Pedro A M Mediano ^{7 8}, Ivan Mindlin ^{6 9}, Giovanni Petri ^{5 10 11},
Daniel Bor ^{12 13}, Fernando E Rosas ^{14 15 16 17}, Antonella Tramacere ¹⁸, Mar Estarellas ^{12 13}



Pairwise analysis



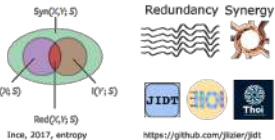
Cabrera, et al, 2017, NeuroImage
Deco, et al, 2015, Nature Reviews Neuroscience
Rubinov, Sporns, 2010, NeuroImage



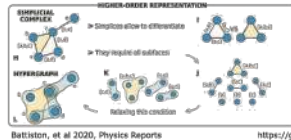
Brain Connectivity Toolbox
brain-connectivity-toolbox.net

High-order analysis

Multivariate Information theory

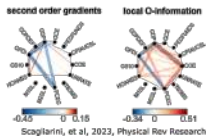


Topology

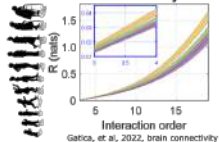


High-order applications

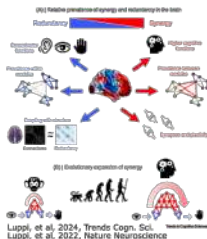
Macroeconomic data



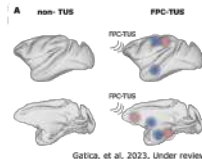
Healthy aging



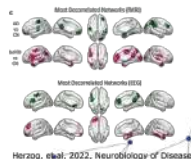
Evolution and cognition



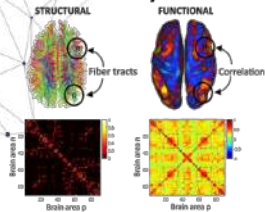
Transcranial ultrasound Stimulation



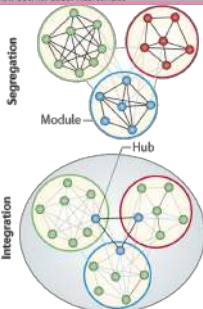
Neurodegeneration



Pairwise analysis



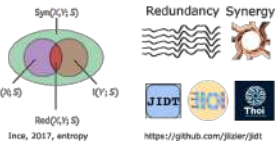
Cabrera, et al, 2017, NeuroImage
Deco, et al, 2015, Nature Reviews Neuroscience
Rubinov, Sporns, 2010, NeuroImage



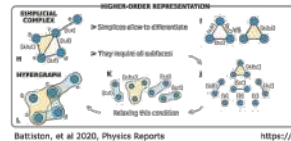
Brain Connectivity Toolbox
brain-connectivity-toolbox.net

High-order analysis

Multivariate Information theory

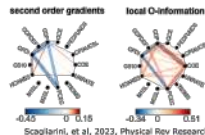


Topology

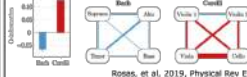


High-order applications

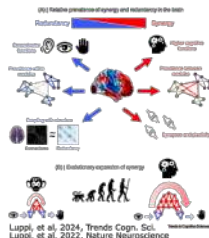
Macroeconomic data



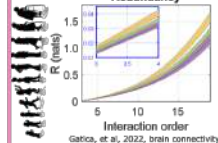
Music



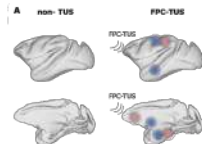
Evolution and cognition



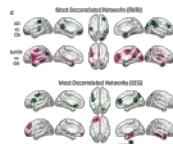
Healthy aging Redundancy

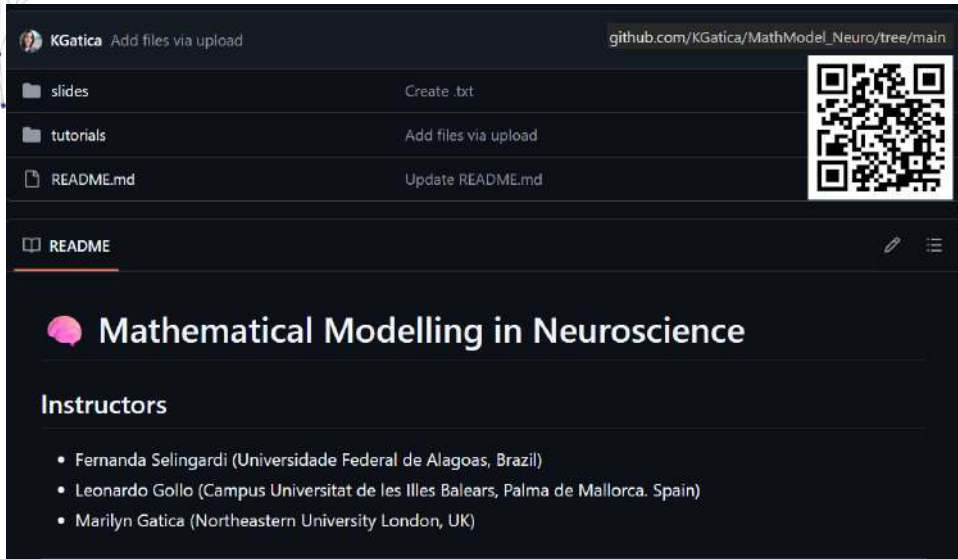


Transcranial ultrasound Stimulation



Neurodegeneration





The screenshot shows a GitHub repository interface. At the top, the repository name 'KGatica' is visible with a profile picture and the text 'Add files via upload'. The URL 'github.com/KGatica/MathModel_Neuro/tree/main' is displayed in the top right. Below the repository name, there are three items: a folder named 'slides' with a 'Create .txt' button, a folder named 'tutorials' with an 'Add files via upload' button, and a file named 'README.md' with an 'Update README.md' button. A QR code is located on the right side of the repository view. Below the repository view, the 'README' file is selected, showing a pink brain icon and the title 'Mathematical Modelling in Neuroscience'. Underneath the title, the section 'Instructors' is listed with three bullet points: 'Fernanda Selingardi (Universidade Federal de Alagoas, Brazil)', 'Leonardo Gollo (Campus Universitat de les Illes Balears, Palma de Mallorca. Spain)', and 'Marilyn Gatica (Northeastern University London, UK)'.

KGatica Add files via upload github.com/KGatica/MathModel_Neuro/tree/main

slides Create .txt

tutorials Add files via upload

README.md Update README.md

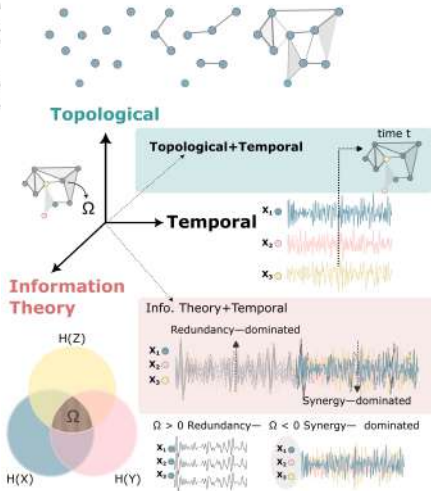
README

 **Mathematical Modelling in Neuroscience**

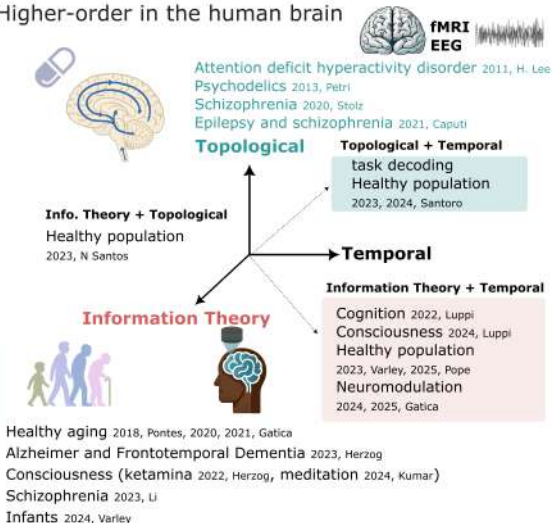
Instructors

- Fernanda Selingardi (Universidade Federal de Alagoas, Brazil)
- Leonardo Gollo (Campus Universitat de les Illes Balears, Palma de Mallorca. Spain)
- Marilyn Gatica (Northeastern University London, UK)

A) High-order interdependencies

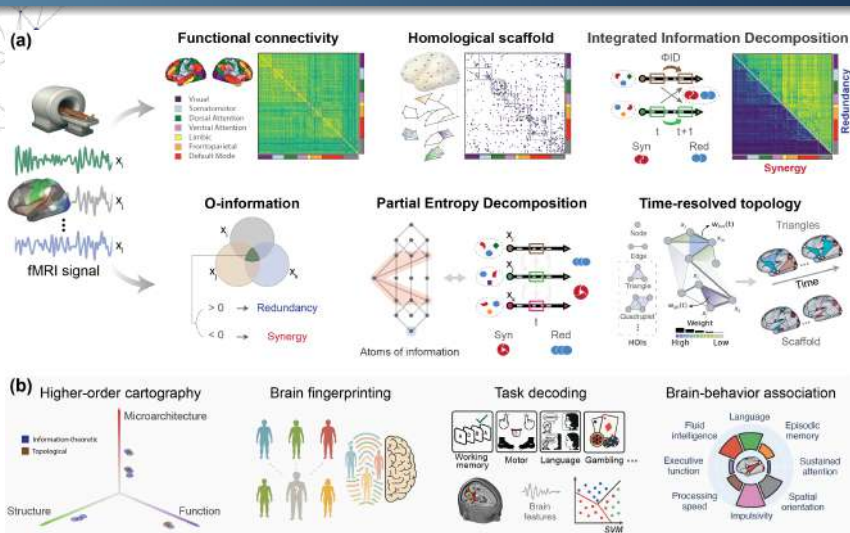


B) Higher-order in the human brain



Carlos Coronel-Oliveros, Marilyn Gatica, Ruben Herzog, Matteo Neri, 2025, Neuroscience

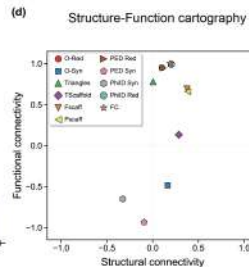
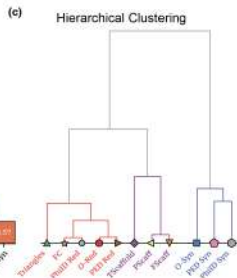
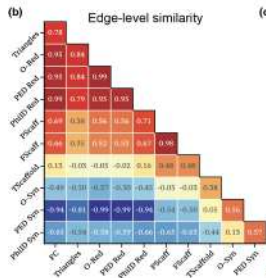
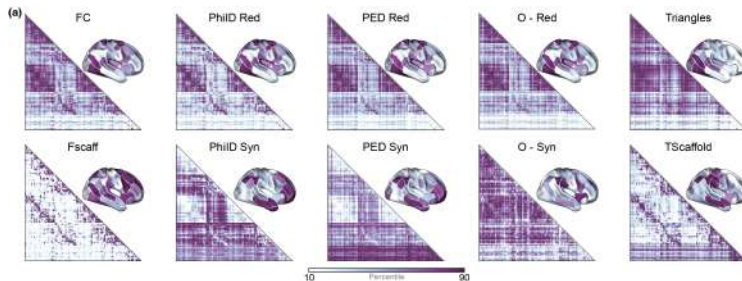
Charting Higher-Order Models of Brain Function



Andrea Santoro,

Matteo Neri, Simone Poetto, Davide Orsenigo, Matteo Diano, Marilyn Gatica, Giovanni Petri, 2025. Under Review

Charting Higher-Order Models of Brain Function



Beyond Pairwise Interactions: Charting Higher-Order Models of Brain Function

Andrea Santoro,^{1,*} Matteo Neri,^{2,*} Simone Poetto,^{3,†} Davide Orsenigo,^{4,1,†}
Matteo Diano,⁴ Marilyn Gatica,^{5,6,‡} and Giovanni Petri^{5,7,1,§}

¹*CENTAI Institute, 10138 Torino, Italy*

²*Institut de Neurosciences de la Timone (INT), UMR 7289,
CNRS, Aix-Marseille Université, Marseille, France*

³*Nicolaus Copernicus University, Toruń, Poland*

⁴*Department of Psychology and Neuroscience Institute of Turin (NIT), University of Turin, Turin, Italy*

⁵*NPLab, Network Science Institute, Northeastern University London, London, E1W 1LP, United Kingdom*

⁶*Precision Imaging, School of Medicine, University of Nottingham, United Kingdom*

⁷*Department of Physics, Northeastern University, Boston, USA*

(Dated: June 24, 2025)



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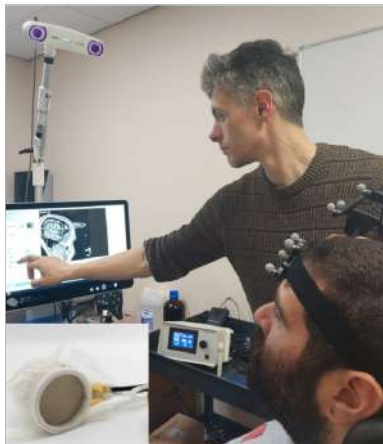
Transcranial ultrasound stimulation



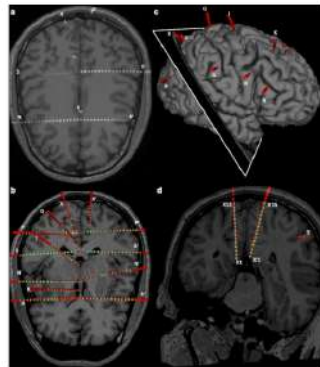
TMS



TUS

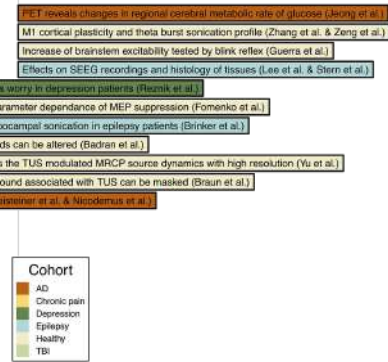


SEEG



Milestones in

Since 2013: <50 human TUS studies medRxiv, 2024, Ca'faratti et al
Research sites: worldwide ~ 120, UK = 5 <https://www.fasfoundation.org/>





Article | [Open access](#) | Published: 15 February 2025

Neuromodulation perception by the general public

[Cyril Atkinson-Clement](#) , [Andrea Junor](#) & [Marcus Kaiser](#)

[Scientific Reports](#) **15**, Article number: 5584 (2025) | [Cite this article](#)

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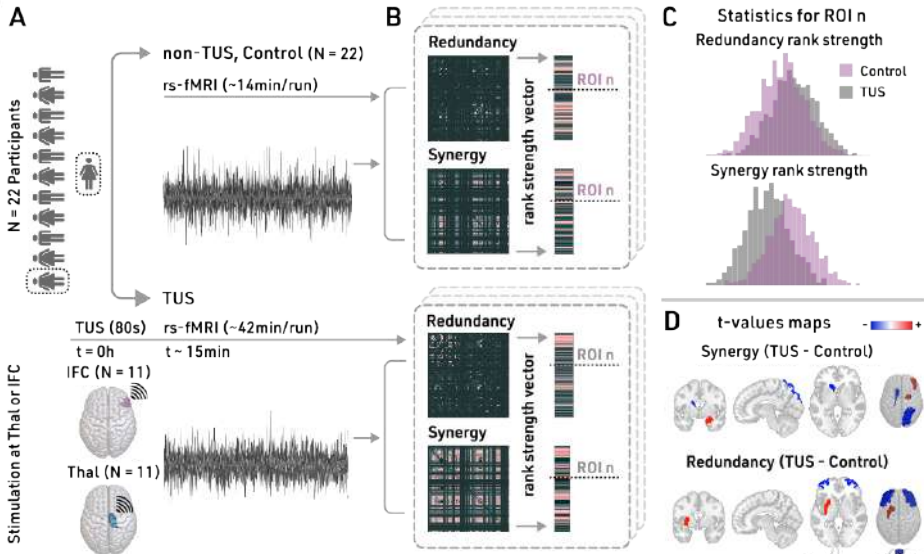
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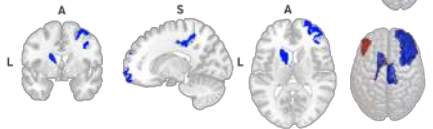


TUS in humans

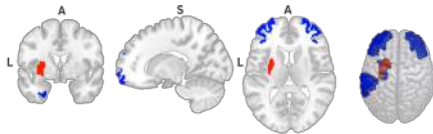


A

Redundancy changes



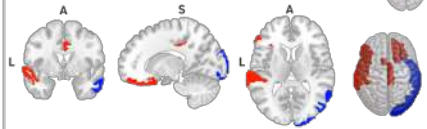
Synergy changes



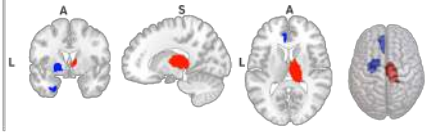
TUS-IFC

B

Redundancy changes



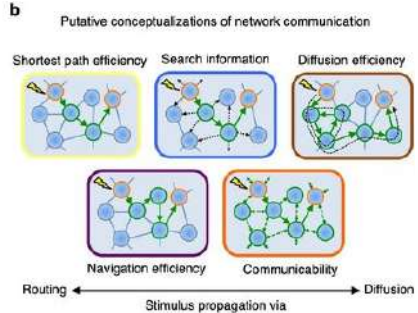
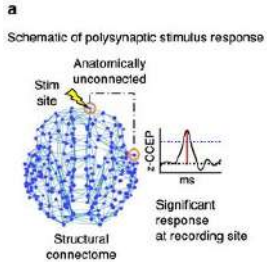
Synergy changes



TUS-Thal

Gatica, et.al, Under review

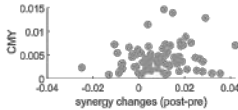
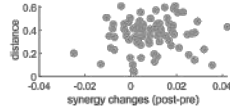
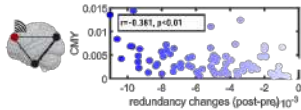
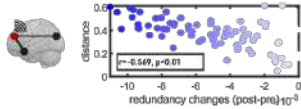
Communication models



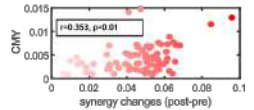
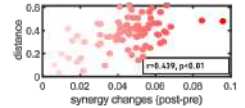
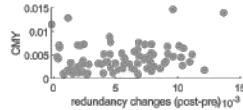
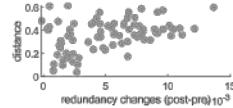
Seguin, et.al, 2023, Neuron

High order changes associated with distance and CMY

A TUS-IFC



B TUS-Thal

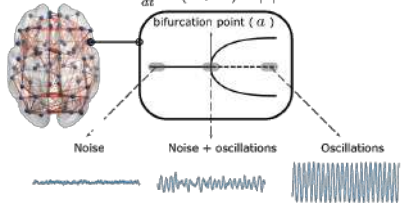


Gatica, et.al, Under review

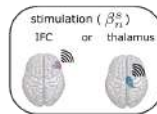
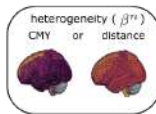
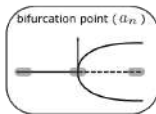
A Whole-brain model

Stuart Landau oscillators (local dynamics)

$$\frac{dz}{dt} = (a + iz) - |z|^2 z$$



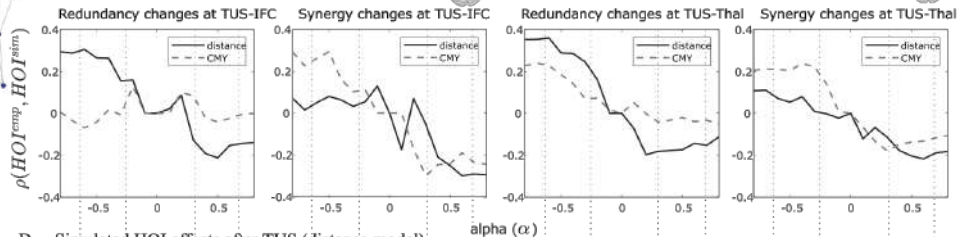
B Regional heterogeneity and stimulation



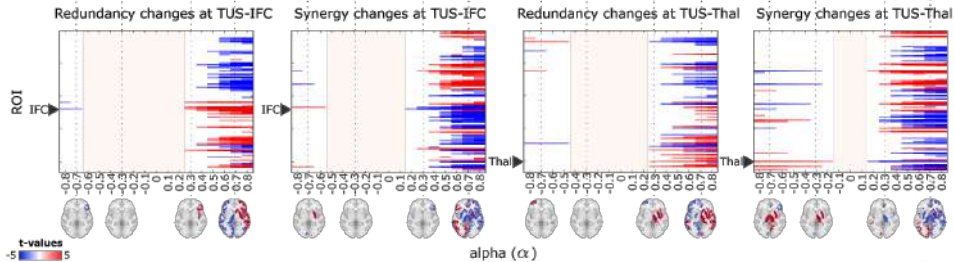
Control: $a_n = \text{bias} + \text{scale } \beta^n$

TUS: $a_n = \text{bias} + \text{scale } (\beta_n + \alpha \beta_n^s)$

C Simulated vs Empirical HOI changes after TUS



D Simulated HOI effects after TUS (distance model)



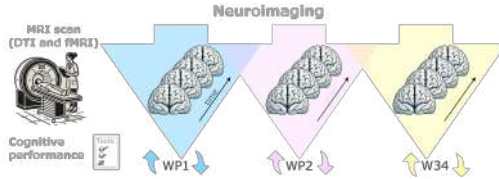
Charting HOI

- **Three HOI families.** Redundancy, synergy, and topological scaffolds. Triangles track redundancy, scaffolds track synergy.
- **Applications diverge.** HOI metrics outperform pairwise FC in brain fingerprinting, while pairwise FC excels in task decoding.

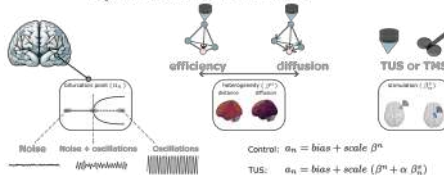
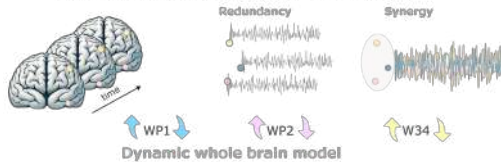
Neuromodulation

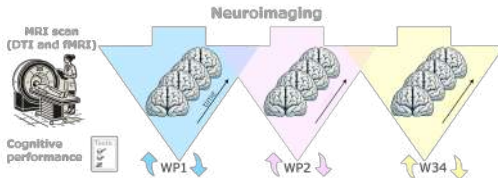
- **HOI as a biomarker of indirect TUS effects.** The TUS produced specific changes depending on the stimulating target.
- This approach opens a potential pathway for predicting changes in brain function following ultrasound stimulation

Towards an integrated framework



Data-drive approach: high-order interactions

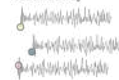




Data-drive approach: high-order interactions



Redundancy



Synergy



Dynamic whole brain model

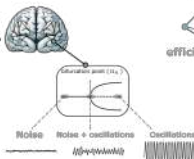


Thank you very much

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Control: $a_n = \text{bias} + \text{scale } \beta^n$
TUS: $a_n = \text{bias} + \text{scale } (\beta^n + \alpha \beta_n^*)$